BioMADE Survey Experiment (2026): GMO/Non-GMO and Products

Data were collected between January 12 and March 3, 2026, via Qualtrics. We received 1,011 completed responses in the raw dataset. The survey contained an embedded experiment that manipulated the definition of biomanufacturing. The 2 (GMO vs. Non-GMO) $\times$ 3 (food, bandages, or footwear) between-subjects experiment contained stimuli that either included or excluded mentions of genetically modified organisms (GMOs), and varied the type of product used as an example.

Participants were randomly assigned to one of the six experimental groups (Table 1). After exposure to the experimental stimulus, we measured participants' emotional reactions to the stimulus, their support for research on biomanufacturing, risk and benefit perceptions, attitudes toward labeling biomanufactured products, and their willingness to pay more for a biomanufactured product.

Emotional reactions to biomanufacturing after exposure to the stimulus

After reading the definition of biomanufacturing, respondents were asked to indicate the extent to which they felt **worried**, **concerned**, **hopeful**, **optimistic**, **interested**, and **curious** (Table 2).

Table 1: Frequency distribution of respondents in experimental groups (N = 1,011).

stim	Frequency	Percentage
GMO / Bandages	177	17.5
GMO / Food	158	15.6
GMO / Footwear	164	16.2
Non-GMO / Bandages	172	17.0
Non-GMO / Food	168	16.6
Non-GMO / Footwear	172	17.0

Table 2: Mean and standard deviation of emotional reactions by experimental condition.

	GMO			Non-GMO			
Characteristic	Bandages	Food	Footwear	Bandages	Food	Footwear	**p-value**[1]
worried							<0.001
Mean	3.56	3.96	3.28	3.46	4.11	3.34	
(SD)	(1.80)	(1.88)	(1.86)	(1.79)	(1.90)	(1.78)	
concerned							0.036
Mean	4.09	4.45	4.04	4.18	4.51	3.94	
(SD)	(1.91)	(1.79)	(1.92)	(1.94)	(1.92)	(2.01)	
hopeful							<0.001
Mean	5.73	4.99	5.39	5.64	4.96	5.49	
(SD)	(1.40)	(1.75)	(1.60)	(1.45)	(1.83)	(1.49)	
optimistic							<0.001
Mean	5.53	4.88	5.13	5.54	4.52	5.46	
(SD)	(1.48)	(1.76)	(1.75)	(1.48)	(2.04)	(1.47)	
interested							<0.001
Mean	5.77	5.03	5.34	5.72	4.95	5.37	
(SD)	(1.58)	(1.93)	(1.81)	(1.44)	(1.82)	(1.80)	
curious							0.003
Mean	5.75	5.26	5.32	5.82	5.36	5.54	
(SD)	(1.49)	(1.81)	(1.83)	(1.33)	(1.76)	(1.62)	

[1]One-way analysis of means (not assuming equal variances)

Table 3: Mean and standard deviation of support and risk and benefit perceptions by experimental condition (N = 1,011).

	GMO			Non-GMO			
Characteristic	Bandages	Food	Footwear	Bandages	Food	Footwear	**p-value**[1]
support							<0.001
Mean	5.49	4.82	5.00	5.22	4.76	4.91	
(SD)	(1.07)	(1.56)	(1.39)	(1.21)	(1.54)	(1.36)	
benefits							0.10
Mean	5.45	4.99	5.20	5.37	5.06	5.02	
(SD)	(1.18)	(1.52)	(1.25)	(1.25)	(1.42)	(1.37)	
risks							0.2
Mean	4.67	5.08	4.73	4.57	4.81	4.53	
(SD)	(1.34)	(1.47)	(1.36)	(1.29)	(1.55)	(1.48)	

[1]One-way analysis of means (not assuming equal variances)

Support for biomanufacturing research

After exposure to the stimulus, we also asked respondents the extent to which they support biomanufacturing research, as well as federal and state funding of biomanufacturing research. We created a single index of support by averaging respondents' scores on these three questions.

In addition to support, we measured risk and benefit perceptions as separate questions. Table 3 shows the mean values and standard deviations of support, and risk and benefit perceptions by experimental condition.

Labeling and willingness to pay more for biomanufactured products

We also included questions (following exposure to the stimulus) that tapped respondents' attitudes toward labeling of biomanufactured products and their willingness to pay more for biomanufactured goods (Table 4).

Table 4: Mean and standard deviation by experimental condition (N = 1,011).

	GMO			Non-GMO			
Characteristic	Bandages	Food	Footwear	Bandages	Food	Footwear	**p-value**[1]
labeling							0.094
Mean	5.98	6.08	5.97	5.43	5.82	5.80	
(SD)	(1.02)	(1.23)	(1.19)	(1.50)	(1.30)	(1.07)	
paymore							0.008
Mean	3.32	2.71	3.17	3.31	2.60	2.66	
(SD)	(1.70)	(1.58)	(1.77)	(1.74)	(1.80)	(1.26)	

[1]One-way analysis of means (not assuming equal variances)